

Gecia Bravo-Hermsdorff

LECTURER/ASSOC PROFESSOR · DEPARTMENT OF COMPUTER SCIENCE @ UNIVERSITY OF BATH
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Education

Princeton University

Princeton, NJ, USA

PHD IN QUANTITATIVE & COMPUTATIONAL NEUROSCIENCE ([link](#))

2020

- *Dissertation*: “Quantifying human priors over abstract relational structures”
- *Selected courses* ([hyperlinked](#)): Random graphs and networks, Mathematical physics, Natural algorithms, Theory of deep learning, Complex analysis, Statistical learning and nonparametric estimation, Machine learning & pattern recognition, Optimal learning, Abstract algebra, Computational complexity, Statistical optimization and reinforcement learning, Stochastic processes on graphs

École Normale Supérieure (ENS Ulm)

Paris, France

RESEARCH MASTER IN COGNITIVE SCIENCES AND NEUROSCIENCE ([link](#))

2011

- *Dissertation*: “Neural basis of self-contingency detection in 5-month-old babies”

DIPLÔME DE L'ENS ([link](#))

2011

- Admitted via the “International Selection in Science” ([link](#))
- Three-year multidisciplinary program, coursework included: Computational neuroscience, Cognitive science, Decision theory, Biophysics, Logic, Mathematics, Statistics, Modeling, Ecology and evolutionary biology, Philosophy of science, Theoretical chemistry

BACHELOR OF SCIENCE ([link](#))

2009

Employment

Dept of Computer Science – University of Bath

Bath, UK

LECTURER (ASSOCIATE PROFESSOR)

Oct 2025–Now

School of Informatics – University of Edinburgh

Edinburgh, Scotland, UK

RESEARCH FELLOW

Oct 2024–Oct 2025

Dept of Statistical Science – University College London

London, UK

RESEARCH FELLOW

2022–2024

- **Causal Graphical Models for Relational Data** ($\geq$ Oct 2023, with Kayvan Sadeghi): Systematically categorizing the conditional independence structures induced by natural models of growing random networks, analyzing their asymptotic distributions, and characterizing the relevant notions of intervention ([link](#)).
- **Extrapolating Interventions** ($\leq$ Oct 2023, with Ricardo Silva): Using our framework of *interventional factor models*, we show when and how data collected under different interventions can be used to characterize the outcome of (unseen) combinations of interventions ([link](#)). We also worked on principled methods for causal-effect estimation using *imperfect instrumental variables* ([link1](#), [link2](#)).

Google Research – Algorithms and Theory

US (remote) and London, UK

AI RESIDENT

2020–2022

- **Privacy**: Led a project that provided a class of data collection protocols that ensure anonymity to users without the need for a fully-trusted central entity performing the anonymization ([link](#)).
- **Geometric Deep Learning**: Developed and implemented (in TensorFlow) a permutation and rotation equivariant neural network architecture for analyzing images of cells from drug discovery experiments, with the aim of identifying compounds that are likely to be effective against the dormant phase of the parasite responsible for Malaria.
- **Graphs**: Implemented (production code in C++) a method for scalable computation of graph cumulants with up to 3 edges [G&B-H 2020 ([link](#))], with the aim of detecting atypical patterns of information propagation in social networks.
- **Entropy**: Developed private and communication efficient algorithms for estimating entropies (work published at NeurIPS, 2022). Implemented the algorithm from [Jian et al 2015 ([link](#))] for computing entropy in the sparse data regime (production code in C++).

Publications

CAUSAL MODELS FOR GROWING NETWORKS [\(link\)](#)

[G Bravo-Hermsdorff](#), K Sadeghi & LM Gunderson. *UAI*, 2025

BUDGETIV: OPTIMAL PARTIAL IDENTIFICATION OF CAUSAL EFFECTS WITH MOSTLY INVALID INSTRUMENTS [\(link\)](#)

J Penn, [G Bravo-Hermsdorff](#), LM Gunderson, R Silva & DS Watson. *AISTATS*, 2025 [\(code\)](#)

BOUNDING CAUSAL EFFECTS WITH LEAKY INSTRUMENTS [\(link\)](#)

DS Watson, J Penn, LM Gunderson, [G Bravo-Hermsdorff](#), A Mastouri & R Silva. *UAI*, 2024 [\(code\)](#)

INTERVENTION GENERALIZATION: A VIEW FROM FACTOR GRAPH MODELS [\(link\)](#)

[G Bravo-Hermsdorff](#), DS Watson, J Yu, J Zeitler & R Silva. *NeurIPS*, 2023 [\(code, poster\)](#)

THE GRAPH PENCIL METHOD: MAPPING SUBGRAPH DENSITIES TO STOCHASTIC BLOCK MODELS [\(link\)](#)

LM Gunderson, [G Bravo-Hermsdorff](#) & P Orbanz. *NeurIPS*, 2023 [\(poster\)](#)

QUANTIFYING NETWORK SIMILARITY USING GRAPH CUMULANTS [\(link\)](#)

[G Bravo-Hermsdorff](#)*, LM Gunderson*, PA Maugis & CE Priebe.

Journal of Machine Learning Research (JMLR), 24(187):1-27, 2023 [\(poster\)](#)

QUANTIFYING HUMAN PRIORS OVER SOCIAL AND NAVIGATION NETWORKS [\(link\)](#)

[G Bravo-Hermsdorff](#). *ICML*, 2023 [\(demo, poster\)](#)

PRIVATE AND COMMUNICATION-EFFICIENT ALGORITHMS FOR ENTROPY ESTIMATION [\(link\)](#)

[G Bravo-Hermsdorff](#), R Busa-Fekete, M Ghavamzadeh, A Muñoz Medina & U Syed. *NeurIPS*, 2022 [\(video\)](#)

STATISTICAL ANONYMITY: QUANTIFYING REIDENTIFICATION RISKS WITHOUT REIDENTIFYING USERS [\(link\)](#)

[G Bravo-Hermsdorff](#), R Busa-Fekete, LM Gunderson, A Muñoz Medina & U Syed. *arXiv*, 2022

INTRODUCING GRAPH CUMULANTS: WHAT IS THE VARIANCE OF YOUR SOCIAL NETWORK? [\(link\)](#)

LM Gunderson* & [G Bravo-Hermsdorff](#)*. *arXiv*, 2020 [\(video, code\)](#)

QUANTIFYING HUMAN PRIORS OVER ABSTRACT RELATIONAL STRUCTURES [\(link\)](#)

[G Bravo-Hermsdorff](#). *Ph.D. dissertation, Princeton University*, 2020 [\(slides, demos\)](#)

A UNIFYING FRAMEWORK FOR SPECTRUM-PRESERVING GRAPH SPARSIFICATION AND COARSENING [\(link\)](#)

[G Bravo-Hermsdorff](#)* & LM Gunderson*. *NeurIPS*, 2019 [\(video, demos, code, poster\)](#)

GENDER AND COLLABORATION PATTERNS IN A TEMPORAL SCIENTIFIC AUTHORSHIP NETWORK [\(link\)](#)

[G Bravo-Hermsdorff](#), V Felso, E Ray, LM Gunderson, ME Helander, J Maria & Y Niv.

Applied Network Science, 4(1), 2019 [\(dataset\)](#)

MODELING THE HEMODYNAMIC RESPONSE FUNCTION FOR PREDICTION ERRORS IN THE VENTRAL STRIATUM [\(link\)](#)

[G Bravo-Hermsdorff](#) & Y Niv. *bioRxiv*, 2019

QUANTIFYING HUMANS' PRIORS OVER GRAPHICAL REPRESENTATIONS OF TASKS [\(link\)](#)

[G Bravo-Hermsdorff](#), TD Pereira & Y Niv.

Unifying Themes in Complex Systems IX. ICCS, Springer Proceedings in Complexity, 281–290, 2018

*denotes equal contribution

Teaching

SPATIO-TEMPORAL ANALYTICS (UNIVERSITY OF BATH) [\(link\)](#)

Winter-Spring 2026

PROBABILITY AND STATISTICS II STAT0005 (UNIVERSITY COLLEGE LONDON) [\(link\)](#)

Fall 2023

- Last mandatory course for bachelors and masters students in the Statistical Science Department, lectures given by Kayvan Sadeghi. Topics included: transformation of random variables, relations between standard distributions, statistical estimation, consistency, method of moments, Bayesian inference, conjugate priors, asymptotic guarantees.
- I held two one-hour tutorial sessions per week covering the students' homework and questions.

BIO MATH BOOTCAMP (PRINCETON UNIVERSITY) [\(link\)](#)

Summer 2016

- Month-long training in mathematical and computational tools for incoming PhD students in the computational neuroscience and computational biology programs, organized by Carlos Brody.
- I lectured for the probability module, and held afternoon sessions for exercises in: programming (Python), linear algebra, ordinary differential equations (ODEs), nonlinear dynamical systems, probability, Fourier transforms, convolutions, and signal processing.

INTRODUCTION TO COGNITIVE NEUROSCIENCE (PRINCETON UNIVERSITY) [\(link\)](#)

Spring 2015

- I held three one-hour sessions per week discussing relevant journal publications, and helped construct and grade the exams.

LAB FOR INTRODUCTION TO EXPERIMENTAL PSYCHOLOGY (PRINCETON UNIVERSITY)

Fall 2014

- I held a weekly three-hour lab session with introductory lectures and exercises in: statistical analysis, MRI, EEG, psychophysics, experimental design, programming, computational modeling, and game theory.

Awards and fellowships

- **TOP REVIEWER AWARD FOR LEARNING ON GRAPHS (LOG) CONFERENCE (PRIZE OF \$1,500)** 2023
- **GOOGLE AI RESIDENCY NYC (ALGORITHMS AND THEORY BRANCH)** 2020–2022
Competitive position for exploring research at Google
- **INDEPENDENT RESEARCH GRANT (\$5,000)** 2019
Funding awarded by the Princeton Cognitive Science Department to selected research proposals
- **SCHOLARSHIP FOR LAKE COMO SCHOOL OF ADVANCED STUDIES IN COMPLEX NETWORKS** May, 2016
- **COGNITIVE SCIENCE GRADUATE FELLOWSHIP** 2016–2017
- **SCHOLARSHIP FOR BRAINS, MINDS AND MACHINES SUMMER SCHOOL** August, 2015
- **SCHOLARSHIP FOR SAMSI BAYESIAN NONPARAMETRICS WORKSHOP** July, 2015
- **PRINCETON PHD FELLOWSHIP** 2013–2019
- **ÉCOLE NORMALE SUPÉRIEURE (ENS ULM) “INTERNATIONAL SELECTION IN SCIENCE”** 2008
- **SCHOLARSHIP BY THE LIONS CLUB TO STUDY FRENCH LITERATURE IN FRANCE** Summer, 2006
- **BRAZILIAN CNPQ “SCIENTIFIC INITIATION” SCHOLARSHIP** 2006–2008
- **ENTRANCE EXAM FOR BIOMEDICAL SCIENCES DEGREE AT THE UNIVERSIDADE FEDERAL DO RIO DE JANEIRO** 2006
Top Brazilian undergraduate program in biomedical sciences, completed 2 of 4 years before moving to France
- **99TH PERCENTILE AT THE EXAME NACIONAL DE ENSINO MÉDIO (ENEM)** 2005
Nationwide exam for Brazilian students after high school
- **TRAVEL AWARDS FOR CONFERENCES AND WORKSHOPS:**
 - Eurandom Workshop on Graph Laplacians, Multivariate Extremes and Algebraic Statistics [\(link\)](#) TU Eindhoven, Netherlands, 2024
 - Foundations of Quantum Computing (FQC2024) Workshop [\(link\)](#) Royal Holloway, University of London, 2024
 - YES Causal Inference Workshop [\(link\)](#) Eurandom, Eindhoven University of Technology, 2023
 - Neural Information Processing Systems (NeurIPS) Scholar Award, 2022 and 2019
 - International Conference on Complex Systems (ICCS), 2018
 - NeurIPS Women in Machine Learning, 2018 and 2017
 - Society for Industrial and Applied Mathematics (SIAM) Annual Meeting, 2018
 - Multidisciplinary Conference in Reinforcement Learning and Decision Making (RLDM), 2017 and 2015
 - International Conference on Mathematical Neuroscience (ICMNS), 2017
 - Austin Memory & Learning Conference, 2015

Academic service and activities

Science outreach

VOLUNTEER JUDGE FOR ADAHACK 2025 [\(link\)](#)

Edinburgh, Oct 2025

- A hackathon organized by *Edinburgh Hoppers*, a society of computer science students for gender minorities at Edinburgh University.

VOLUNTEER JUDGE FOR PRINCETON RESEARCH DAY 2025 [\(link\)](#)

Online, April 2025

VOLUNTEER TEACHER FOR THE IN2STEM-IN2SCIENCEUK OUTREACH INITIATIVE [\(link\)](#)

London, August 2024

- Designed and taught three 2-hour classes to nine high-school students:
 - The mathematics of card magic tricks* — based on the wonderful book “Magical Mathematics” by Diaconis and Graham [\(link\)](#);
 - Optimal betting* — derive the Kelly criterion [\(link\)](#) using hands-on simulations and a story about exploiting a broken arcade game;
 - The mathematics of cooperation* — a friendly introduction to evolutionary game theory using as entry point the classical work by Axelrod on tournaments of the repeated Prisoner’s Dilemma game [\(link\)](#).
- Our placement (which was hosted by Alex Watson) was picked as “host of the week” thanks to nominations from our students [\(link\)](#).

PEDAGOGY TRAINING

- One day workshop by the Alda Center for Communicating Science [\(link\)](#).

Princeton, October 2018

VOLUNTEER AT THE PRINCETON NEUROSCIENCE FAIR

Princeton, March 2018

- Event with fun neuroscience demonstrations for 4th grade of low-income households from the Christina Seix Academy.

SCIENCE OUTREACH VOLUNTEER AT STEMCIVICS CHARTER SCHOOL

Ewing, NJ, April 2017

- I performed a demonstration of a cerebellar illusion using glasses that distort ones aim as they throw a ball.

Invited workshops

GRAPH LAPLACIANS, MULTIVARIATE EXTREMES AND ALGEBRAIC STATISTICS [\(link\)](#)

Eurandom, Nov 2024

FOUNDATIONS OF QUANTUM COMPUTING (FQC2024) WORKSHOP [\(link\)](#)

Royal Holloway, University of London, August 2024

NETWORKS AND TIME MEETING II [\(link\)](#)

Northeastern University London, April 2024

YES CAUSAL INFERENCE WORKSHOP [\(link\)](#)

Eurandom, Eindhoven University of Technology, March 2023

Mentoring

GRADUATE

- Katie Robertson — *Master student at University of Edinburgh* Feb 2025–Sep 2025
I proposed and supervised her Msc thesis project on efficiently counting medium-sized subgraphs.
- Jordan Penn — *PhD student at King’s University College* Winter 2023–now
I helped supervise his project on analyzing phase transitions in dynamical network models and his projects on methods for causal inference in the presence of unobserved confounders.
- Emma Graham (now PhD student at Dartmouth) — *Master’s student at UCL Center for Artificial Intelligence* Summer 2023
I helped supervise her MSc thesis project on using Voronoi diagrams to study catastrophic forgetting in reinforcement learning.

UNDERGRADUATE

- Cristian Andronic — *Computer science major at Princeton University* 2016
I proposed and supervised a project to build a navigation App to collect data for modeling naturalistic human mobility.
- Daniel J. Wilson (now PhD student at University of Toronto) — *Volunteer intern at Princeton Neuroscience Institute (PNI)* 2015
I proposed and supervised a project involving coding a contextual bandit task in MTurk [\(link\)](#) to model human representation learning.
- Caitlyn Cap and Olamilekan Sule — *Summer interns at the Botvinick lab on PNI* 2014

Hackathons

WEEKEND-LONG MIT QUANTUM HACKATHON (IQUHACK) [\(link\)](#)

Remote, Feb 2024

- I worked on the IonQ challenge to reverse-engineer quantum circuits [\(link\)](#).

DAY-LONG BIOHACKATHON BY PRECISIONLIFE AND ME RESEARCH UK [\(link\)](#)

UCL, Feb 2024

- Exploratory analysis of a large database of single nucleotide polymorphisms (SNPs) correlated with Myalgic Encephalomyelitis (ME).

WEEK-LONG PNI-INTEL HACKATHON ON MULTIVARIATE ANALYSIS OF FMRI DATA

Princeton, Jan 2017

Reviewer

JOURNALS

- Network Science (2024); IEEE Transactions on Signal Processing (2024); Scandinavian Journal of Statistics (2024); Socio-Economic Planning Sciences (2019); Trends in Cognitive Sciences (2017)

CONFERENCES

- Conference on Uncertainty in Artificial Intelligence (UAI) (2026); Conference on Neural Information Processing Systems (NeurIPS) (2025, 2022); Learning on Graphs Conference (LoG) (2023, 2024); International Conference on Machine Learning (ICML) (2022, 2023); TheWebConf2023; WHMD 2021 NeurIPS workshop; NeurIPS Women in Machine Learning (2018, 2017)

Summer schools

IX PARATY QUANTUM INFORMATION SCHOOL [\(link\)](#)

QUANTUM PARATY

August 2025

Paraty, Brazil

SUMMER SCHOOL ON MATHEMATICAL ASPECTS OF QUANTUM INFORMATION [\(link\)](#)

UNIVERSITÉ PARIS-SACLAY AND INSTITUT POLYTECHNIQUE DE PARIS

June, 2025

Paris, France

QUANTUM MACHINE LEARNING AND HAMILTONIAN SIMULATION [\(link\)](#)

LONDON MATHEMATICAL SOCIETY (LMS)

June 2025

Isle of Skye, Scotland

MACHINE LEARNING SUMMER SCHOOL (MLSS) [\(link, my video ~9min\)](#)

MAX PLANCK INSTITUTE FOR INTELLIGENT SYSTEMS

Summer 2020

Tübingen, Germany (virtual)

COMPLEX NETWORKS: THEORY, METHODS, AND APPLICATIONS [\(link\)](#)

LAKE COMO SCHOOL OF ADVANCED STUDIES

May 2016

Lake Como, Italy

BRAINS, MINDS AND MACHINES (BMM) SUMMER COURSE [\(link\)](#)

MIT CENTER FOR BRAINS, MINDS AND MACHINES

August 2015

Woods Hole, MA, USA

BAYESIAN NONPARAMETRICS: SYNERGIES BETWEEN STATISTICS, PROB AND MATH [\(link\)](#)

STATISTICAL AND APPLIED MATHEMATICAL SCIENCES INSTITUTE (SAMSI)

June 2015

Durham, NC, USA

COMPUTATIONAL AND COGNITIVE NEUROBIOLOGY SUMMER SCHOOL [\(link\)](#)

COLD SPRING HARBOR LABORATORY ASIA

July 2010

Suzhou, China

Languages

- **Human:** (BRAZILIAN) PORTUGUESE (*native*), ENGLISH & FRENCH (*fluent*), SPANISH (*basic*)
- **Computer:** PYTHON & MATLAB (*fluent*), MATHEMATICA, QISKIT, C++ & R (*functional*), JAVASCRIPT & HTML (*basic*)

Other research experiences

PHD CANDIDATE AT THE NIV LAB [\(link\)](#)

Princeton University, 2014–2019

- I developed methods to efficiently quantify human priors over relational data by exploiting the relevant underlying symmetry [\(link\)](#).

PHD RESEARCH ROTATION AT THE BOTVINICK LAB

Princeton University, 2013–2014

RESEARCH SCHOLAR IN NEUROECONOMICS AT THE MONTAGUE LAB [\(link\)](#)

Virginia Tech, 2011–2013

- I developed and validated computational models to explain human behavioral data from various neuroeconomic experiments, such as, multi-armed bandit tasks and the repeated ultimatum game. *Advisors:* Read Montague and Terry Lohrenz

MASTER'S STUDENT AT THE COGNITIVE SCIENCE AND PSYCHOLINGUISTIC LAB [\(link\)](#)

ENS Ulm, Paris, 2011

- Researched the neural substrates of self-contingency detection in babies using functional near-infrared spectroscopy (fNIRS). I designed, coded, and built the experimental apparatus, and recorded and analyzed data from 61 babies. *Advisor:* Emmanuel Dupoux

RESEARCH INTERNSHIP AT CALTECH EMOTION AND SOCIAL COGNITION LAB [\(link\)](#)

Caltech, Pasadena, Spring 2010

- I designed and carried out behavioral experiments to analyze whether humans express values learned via Pavlovian conditioning in an unrelated task without their conscious awareness. *Advisors:* Naotsugu Tsuchiya and Ralph Adolphs

RESEARCH INTERNSHIP AT THE DEVELOPMENT AND NEUROPHARMACOLOGY LAB [\(link\)](#)

Collège de France, Paris, 2009

- Researched the molecular mechanisms involved in the emergence of cellular territories during the morphogenesis of the neural tube. *Advisors:* Elizabeth Di Lullo and Alain Prochiantz

UNDERGRADUATE STUDENT AT THE PHYSIOLOGY OF COGNITION LAB [\(link\)](#)

UFRJ, Brazil, 2007–2008

- Studied the physiology of the visual system in monkeys (using intracranial recordings) and humans (using EEG). *Advisor:* Mário Fiorani

RESEARCH INTERNSHIP AT THE INSTITUTE OF NEUROBIOLOGY ALFRED FESSARD [\(link\)](#)

CNRS, Gif-sur-Yvette, Summer 2007

- Researched the development of the neural crest by grafting quail and chick embryos *in ovo*. *Advisors:* Sophie Creuzet

Selected talks

- **TO CONTRACT, OR TO DELETE, THAT IS THE QUESTION**

Thirteenth Meeting of the Southern Logic Seminar, University of Bath, Feb 2026 [\(link\)](#) [\(pdf\)](#)

- **CAUSAL MODELS FOR GROWING NETWORK**

Royal Statistics Society (RSS) International Conference, Edinburgh, Sept 2025 [\(link\)](#)

- **WHAT IS THE VARIANCE (AND SKEW, KURTOSIS, ETC) OF A NETWORK?**

GRAPH CUMULANTS FOR NETWORK ANALYSIS

Networks Seminar Series, Mathematical Institute, University of Oxford, Dec 2024 [\(link\)](#)

- **STOCHASTIC GRAPH REDUCTION VIA COMBINATORIAL INVARIANTS [\(link\)](#)**

Eurandom Workshop on Graph Laplacians, Multivariate Extremes and Algebraic Statistics, Eindhoven, Nov 2024

- **PRINCIPLED PROCESSING OF RELATIONAL DATA**

University of Southampton, School of Mathematics, July 2024

- **WHAT IS THE VARIANCE (AND SKEW, KURTOSIS, ETC) OF A NETWORK?**

INTRODUCING GRAPH CUMULANTS FOR NETWORK ANALYSIS

University of Warwick, Department of Computer Science, May 2024

- **A TAXONOMY OF CAUSAL MODELS FOR GROWING NETWORKS**

Northeastern University London Meeting on Networks and Time II, April 2024 [\(link\)](#)

- **CUMULANTS FOR NETWORKS**

Algebraic and Combinatorial Perspectives in the Mathematical Sciences (ACPMS) Seminar, Online, 2022 [\(link\)](#)

- **GRAPH CUMULANTS: WHAT IS THE VARIANCE OF YOUR SOCIAL NETWORK?**

Learning with Graphs Summit (Google), 2022

- **STATISTICAL ANONYMITY: QUANTIFYING REIDENTIFICATION RISK WITHOUT REIDENTIFYING USERS.**

Chrome Privacy Budget Meeting, 2021

- **USING GRAPH CUMULANTS TO DETECT ATYPICAL PATTERNS OF INFORMATION SPREAD IN SOCIAL NETWORKS**

MML Eng Meeting (Google), 2021

- **ENTROPY ESTIMATION OF HIGH-DIMENSIONAL SPARSE DATASETS.**

Chrome Privacy Budget Meeting, 2021

- **GRAPH REDUCTION BY EDGE DELETION AND EDGE CONTRACTION**

International Conference on Complex Systems (ICCS), Cambridge, MA, 2018

- **QUANTIFYING PEOPLE'S PRIORS OVER GRAPHICAL REPRESENTATIONS OF TASKS**

International Conference on Complex Systems (ICCS), Cambridge, MA, 2018

- **GRAPH REDUCTION BY EDGE DELETION AND EDGE CONTRACTION.**

SIAM Workshop on Network Science (SIAMNS18), Portland, Oregon, 2018

- **ASSESSING DECISION-MAKING IN PATIENTS WITH INSULA LESION USING VARIOUS NEUROECONOMIC TASKS**

Regional Conference in Neuroeconomics at the Duke Center for Interdisciplinary Decision Sciences, 2016

Selected publications

QUANTIFYING NETWORK SIMILARITY USING GRAPH CUMULANTS [\(link\)](#)

[G Bravo-Hermsdorff](#)^{*}, LM Gunderson^{*}, PA Maugis & CE Priebe.

Journal of Machine Learning Research (JMLR), 24(187):1-27, 2023 [\(video, code, poster\)](#)

A UNIFYING FRAMEWORK FOR SPECTRUM-PRESERVING GRAPH SPARSIFICATION AND COARSENING [\(link\)](#)

[G Bravo-Hermsdorff](#)^{*} & LM Gunderson^{*}. *NeurIPS*, 2019 [\(video, demos, code, poster\)](#)

CAUSAL MODELS FOR GROWING NETWORKS [\(link\)](#)

[G Bravo-Hermsdorff](#), K Sadeghi & LM Gunderson. *UAI*, 2025

QUANTIFYING HUMAN PRIORS OVER SOCIAL AND NAVIGATION NETWORKS [\(link\)](#)

[G Bravo-Hermsdorff](#). *ICML*, 2023 [\(demo, poster\)](#)